

Yazeed Hasan

End-to-End AI & Data Science Expert | AI/ML Engineering, Data Platforms, MLOps & Delivery Leadership
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Professional Summary

End-to-end AI and Data Science expert with 9+ years in technology, including 7+ years delivering production AI, machine-learning, analytics, and data-platform solutions. Hands-on across business problem framing, data engineering, feature engineering, modeling, generative and agentic AI, MLOps, deployment, monitoring, governance, and continuous optimization.

Combines deep technical execution with leadership of a seven-member Data and AI team, project and portfolio delivery, executive stakeholder management, solution strategy, and commercial alignment. Delivered measurable improvements including 25% higher platform reliability, 20% greater throughput, 40% faster deployments, and 60% lower manual operational workload across telecom, enterprise AI, public-sector analytics, KYC/AML, education, and workflow automation.

Languages

- **Arabic:** Native
- **English:** Professional working proficiency

Interests

- **Professional:** Applied AI, product strategy, open-source technologies, data-driven decision-making
 - **Personal:** Travel, history, chess, walking, film, gaming, exploration
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Certifications

- **Project Management Professional (PMP)** – Project Management Institute (PMI) (Jan 2026)
- **Certified International Trainer in Data Science and Artificial Intelligence** – Board | Oxford College (2024)
- **Professional Machine Learning Engineer** – Google Cloud (Sep 2023)
- **Cloud Digital Leader** – Google Cloud (Aug 2023)
- **TensorFlow Developer Certificate** – TensorFlow Certificate Program (Sep 2022)
- **IELTS – International English Language Testing System** (Mar 2021)
- **PCAP – Certified Associate in Python Programming** (Jan 2020)
- **PCEP – Certified Entry-Level Python Programmer** (Dec 2019)
- **Oracle Certified Professional, Java SE 8 Programmer** (Jul 2017)
- **Oracle Certified Associate, Java SE 8 Programmer** (Feb 2017)

Awards & Recognition

- **Responsible Security Disclosure Recognition** – Google client engagement (2026): identified and responsibly reported security vulnerabilities that informed remediation and strengthened security controls.
 - **Two-Time Best Employee of the Year** – LigaData (2023, 2024): recognized for delivery excellence, technical ownership, and operational impact.
 - **ISO 21001:2018 Certification-Readiness Contribution** – HTU (2019–2020): contributed to training-quality processes, delivery consistency, and accreditation readiness for a non-formal education upskilling program.
 - **Ranked Second in Graduating Class** – The Hashemite University (2019).
 - **Top Students / Best Performers Board Recognition** – The Hashemite University (2017, 2018): recognized twice for academic performance.
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Education

- **B.Sc. in Computer Information Systems** – The Hashemite University, Jordan
2015–2019 | GPA: 3.43/4.00
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Papers & Publications

Towards a Direct Mental Based Decision Making: Electroencephalography (EEG) Case Study
IEEE ICICS 2020 **Published Apr 2020**

- Co-authored the paper published at the 2020 11th International Conference on Information and Communication Systems (ICICS); research and implementation were completed from Jan–Apr 2019.
 - Designed and implemented an AI-enabled brain-computer interface Tetris prototype controlled through EEG and EMG signals, covering signal capture, numerical feature extraction, model inference, and real-time decision mapping.
 - Integrated model outputs with game-control APIs and released the reproducible implementation through the GitHub code repository. DOI: 10.1109/ICICS49469.2020.239502.
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Core Expertise

- **AI, Machine Learning & Data Science:** business problem framing, exploratory analysis, feature engineering, predictive modeling, classification, segmentation, survival analysis, forecasting, experimentation, causal inference, model evaluation, and optimization using Python, Pandas, scikit-learn, XGBoost, and TensorFlow.
- **Generative AI, Agentic Systems & NLP:** LLM applications, RAG, vector search, prompt and context engineering, multi-agent orchestration, tool calling, memory and state patterns, document intelligence, multimodal processing, Text-to-SQL, SQL-to-Text, guardrails, and evaluation using Gemini and Hugging Face technologies.

- **Data Engineering & Distributed Platforms:** Spark, PySpark, Kafka, Airflow, Hadoop, HDFS, BigQuery, BigLake, Dataplex, Dataflow, Dataproc, Trino, Presto, PostgreSQL, Redshift, Redis, ETL/ELT, batch and streaming processing, data quality, and lineage.
- **MLOps, Cloud & Production Architecture:** GCP, AWS, Vertex AI, SageMaker, MLflow, Docker, Kubernetes, Terraform, CI/CD, infrastructure as code, model deployment, monitoring, observability, SLIs/SLOs, incident response, and cost controls.
- **Programming, APIs, Analytics & BI:** Python, SQL, Java, R, Bash, Git, FastAPI, REST APIs, webhooks, enterprise integrations, Looker, Power BI, Amazon QuickSight, Grafana, semantic layers, KPI design, and executive analytics.
- **Technical, Project, Business & Governance Leadership:** team leadership, architecture and code reviews, roadmaps, scope, scheduling, risk, quality, resource prioritization, stakeholder and client management, solution strategy, trade-off analysis, value realization, IAM, RBAC, VPC-SC, DLP, CMEK, privacy, auditability, and compliance.

Professional Experience

Primary employment is identified separately from concurrent contract, freelance, part-time, and internship engagements; overlapping dates reflect intentional parallel work.

Technical Team Lead – AI & Data Technical Leadership LigaData Inc.

Feb 2024–Present

Primary, Full-Time / Hybrid

- Lead end-to-end delivery of enterprise AI, machine-learning, analytics, and data-platform initiatives, translating business objectives into solution architectures, delivery roadmaps, milestones, risks, KPIs, and production outcomes.
- Manage and mentor a seven-member cross-functional Data and AI engineering team across discovery, data engineering, model development, validation, deployment, production support, and continuous improvement.
- Direct architecture and implementation of batch and near-real-time data pipelines, predictive models, LLM/RAG services, agentic workflows, analytics APIs, and executive reporting capabilities.
- Improved platform reliability by 25% and throughput by 20% through performance engineering, pipeline optimization, observability, data-quality controls, and disciplined incident management.
- Reduced deployment time by 40% by standardizing CI/CD, configuration management, automated validation, release governance, and production-readiness controls for data and AI services.
- Establish governance for data quality, lineage, model evaluation, monitoring, documentation, security, and auditability, strengthening trust, compliance, maintainability, and business continuity.
- Partner with executives, customers, and technical stakeholders to shape AI/data strategy, prioritize opportunities, define scope and acceptance criteria, communicate trade-offs, and convert technical capabilities into measurable business value.
- Own architecture and code reviews, resource prioritization, delivery escalations, stakeholder reporting, and technical coaching while aligning engineering execution with project constraints and commercial objectives.

Senior AI Engineer
Alphabet / Google via Petra Pathway Consulting

Oct 2025–Jun 2026

Independent Contract / Remote

- Delivered secure end-to-end AI and data-science solutions on Google Cloud Platform, covering business discovery, data ingestion, feature engineering, modeling, evaluation, deployment, monitoring, and stakeholder adoption.
- Architected scalable lakehouse and AI foundations using BigQuery, BigLake, Dataplex, Dataflow, Dataproc, and Pub/Sub for batch, streaming, classical ML, and generative-AI workloads across enterprise-scale datasets.
- Designed and built a multi-agent AI platform for enterprise knowledge workflows, integrating orchestration, tool calling, memory-management patterns, context compaction, and governed access to structured and unstructured data.
- Engineered multimodal document-intelligence pipelines for PDF and PPTX image extraction, OCR, parsing, metadata enrichment, indexing, source-grounded Q&A, research synthesis, and executive insight generation.
- Developed Gemini-powered Text-to-SQL and SQL-to-Text analytics capabilities with schema grounding, prompt routing, response validation, safety guardrails, conversational context, and governed BigQuery access.
- Built orchestrated pipelines with data contracts, SLIs/SLOs, observability, and cost controls, improving throughput by 35% and reducing latency by 30% while preserving reproducibility and auditability.
- Productionized models on Vertex AI with offline and online evaluation, CI/CD, infrastructure as code, automated validation, model monitoring, and incident response; applied experimentation and causal inference to support product and platform decisions.
- Built BigQuery marts, semantic layers, and Looker dashboards for executive analytics while enforcing least-privilege IAM, VPC-SC, DLP, CMEK, lineage, and secret-management controls.

Lead Data & AI Instructor
StepByStep Academy

Jul 2019–Present

Part-Time / Hybrid

- Lead the design and delivery of advanced Data and AI programs covering Python, SQL, APIs, data engineering, machine learning, deep learning, generative AI, RAG, MLOps, and applied analytics.
- Build end-to-end learning paths that connect business problem framing and data preparation with feature engineering, model training, evaluation, deployment, monitoring, and AI-powered automation.
- Develop project-based curricula using real-world datasets, structured labs, coding exercises, capstone projects, assessments, and production-oriented use cases aligned with industry requirements.
- Mentor learners through architecture decisions, experimentation, debugging, code review, model evaluation, deployment, portfolio development, and career-oriented technical execution.
- Coach instructors and govern curriculum quality by reviewing technical materials, standardizing delivery practices, and aligning enterprise-upskilling outcomes with workforce and business needs.

Big Data Engineer
LigaData Inc.

Jun 2021–Feb 2024

Primary, Full-Time / Hybrid

- Architected telecom-scale data and feature pipelines across Hadoop data lakes, Kafka streams, Spark and PySpark processing, Trino/Presto query layers, Redis, FastAPI, and PostgreSQL.
- Built batch and near-real-time ETL/ELT workflows that transformed, enriched, validated, and served high-volume data for analytics, reporting, machine-learning training, and operational decision support.
- Developed classification, customer-segmentation, and GIS-based machine-learning models using Python and distributed data-processing frameworks to support business decisions and service-impact reduction.
- Engineered Kafka-based event ingestion and Spark processing patterns for reliable streaming analytics, feature preparation, anomaly detection, and downstream data products.
- Exposed curated datasets, analytics outputs, and model results through FastAPI services and Redis-backed low-latency access patterns for downstream applications and stakeholders.
- Built data-quality controls, monitoring dashboards, and alerting across batch and streaming sources to improve freshness, accuracy, observability, and analytics readiness.
- Optimized SQL queries, Spark jobs, Kafka consumers, caching, and transformation logic to improve scalability, processing efficiency, and reliability across production workloads.

Data Scientist
World Bank Group

Mar 2022–Sep 2022

Independent Contract / Remote

- Built a national-scale unemployment analytics pipeline for Jordan by integrating administrative datasets from social security, tax, and civil-registry sources.
- Designed data-ingestion, cleaning, transformation, validation, reconciliation, exploratory-analysis, and feature-engineering workflows to produce reliable analytical and modeling datasets.
- Applied survival analysis and machine-learning classification to estimate employment-risk patterns and support evidence-based labor-market interventions.
- Developed REST APIs to expose model predictions and analytical outputs for automated scoring, reporting, and operational decision support.
- Automated pipeline execution, model retraining, monitoring, and alerting with Airflow while maintaining traceability, reproducibility, and structured data stores.
- Published reproducible project code on GitHub to support transparent, maintainable, and repeatable analytical delivery.

Data Science Instructor
Multiple Organizations

Jul 2019–Sep 2022

Part-Time / Hybrid

- Delivered applied data-science and machine-learning training for universities, academies, and professional providers across Python, R, SQL, statistics, AWS, Power BI, data visualization, and model evaluation.
- Designed hands-on workflows covering data cleaning, exploratory analysis, feature engineering, model development, validation, interpretation, reporting, and deployment fundamentals.
- Created project-based materials, labs, assignments, and real-world datasets that required learners to build, evaluate, and communicate complete data products.
- Mentored students and early-career professionals through technical problem-solving, code review, debugging, portfolio development, and applied AI/ML projects.

Big Data Operations Engineer
LigaData Inc.

Jan 2020–Jun 2021

Primary, Full-Time / Hybrid

- Operated and analyzed telecom-scale Hadoop, Kafka, scheduled-processing, and reporting environments that supplied trusted data to analytics, machine-learning, and business-intelligence use cases.
- Automated recurring data operations, validation, reconciliation, and monitoring using Python, SQL, and shell scripting, reducing manual workload by 60%.
- Monitored ingestion quality, Kafka consumer health, lag, pipeline execution, data freshness, and model-data dependencies to protect service continuity and analytical reliability.
- Investigated anomalies, data drops, delays, and inconsistencies through root-cause analysis, data profiling, reconciliation, pipeline recovery, and preventive controls.
- Produced operational analytics, dashboards, alerts, and reports that translated platform behavior into actionable service and business decisions.
- Worked with business clients and technical stakeholders to capture requirements, validate outputs, prioritize issues, and convert operational needs into reliable data products.

AI Developer
KADDB

Jan 2019–Apr 2019

Project Contract / On-Site

- Developed computer-vision models for real-time object detection, recognition, and tracking in AI-enabled monitoring and visual-analysis scenarios.
- Built data-collection, labeling, preprocessing, model-testing, evaluation, and deployment workflows to improve recognition quality and operational stability.
- Deployed and optimized edge-AI inference on NVIDIA Jetson Nano and Raspberry Pi devices for low-latency execution under constrained compute resources.
- Maintained structured stores for model inputs, outputs, metadata, and operational records, protecting data integrity and reproducibility across the development lifecycle.
- Collaborated with technical stakeholders to validate requirements, test prototypes, analyze results, and adapt models for practical deployment conditions.

Games & Virtual Reality Developer Intern
KADDB

Jan 2019–Apr 2019

Concurrent Internship Assignment / On-Site

- Developed a Unity-based immersive training simulation integrating real-time 3D environments, event-driven interaction logic, and structured simulation workflows.
- Integrated Emotiv Epoc+ EEG signals, OptiTrack motion-tracking data, and holographic-display components to support multimodal human-computer interaction.
- Built C# components for signal and input processing, scene control, event handling, interaction behavior, and experimental data flow.
- Supported testing, calibration, and iterative analysis of VR and sensor components to improve realism, usability, and training effectiveness.

Trainee Java Developer Tuned Application

Nov 2016–Feb 2017

Traineeship / On-Site

- Developed Java SE 8 and JavaFX desktop applications using object-oriented design, modular architecture, event-driven logic, and responsive user interfaces.
- Implemented relational-database interactions, CRUD workflows, validation, file processing, and multithreading, establishing a strong foundation in data-centric software engineering.
- Supported testing, debugging, refactoring, version control, and documentation to improve application reliability, maintainability, and usability.

Professional AI & Data Projects

LigoFlow.ai – AI Project Execution Platform MentoraX Inc.

Sep 2025–Present

- Lead the end-to-end design and engineering of an AI-powered workflow-automation and conversation-intelligence platform that converts fragmented workplace discussions into structured, accountable, and trackable execution.
- Build the AI analysis layer for commitment detection, owner identification, deadline extraction, contextual interpretation, prioritization, and conversion of unstructured conversations into actionable work items.
- Integrate 11+ collaboration, communication, ticketing, and project-management systems, including Slack, Microsoft Teams, Zoho, Jira, Asana, Confluence, GitHub, GitLab, Gmail, Outlook, and Telegram.
- Implement automated ticket creation, task routing, follow-ups, escalations, ownership assignment, SLA tracking, audit logging, and execution-status reporting.
- Architect API-first and webhook-driven services with secure processing, configurable customer environments, private-cloud readiness, on-premises deployment support, and data-sovereignty controls.
- Deliver leadership analytics covering commitments, overdue work, bottlenecks, accountability, collaboration efficiency, risk, and overall project health.

AI Agents Factory Platform LigaData Inc.

Jan 2026–Present

- Lead integration of multiple open-source agent frameworks and orchestration technologies into a unified internal platform for rapid enterprise-agent development.
- Build reusable patterns for multi-agent orchestration, RAG, tool calling, workflow execution, knowledge access, state and memory management, and operational traceability.
- Design authentication, RBAC, auditing, monitoring, logging, governance, and production controls for secure enterprise adoption.
- Productize shared platform capabilities to shorten delivery cycles across internal initiatives and client-facing AI solutions.

CAP Hotline Multi-Agent AI Assistant**Alphabet / Google via Petra Pathway Consulting****Apr 2026–Jun 2026**

- Built a CRM-focused AI assistant that orchestrated specialized agents to answer customer requests using RAG, inventory data, and historical support-ticket analysis.
- Designed conversational workflows for intake, onboarding, issue classification, retrieval, answer generation, guided resolution, and human escalation.
- Integrated automated ticket creation, triage, routing, and escalation into enterprise systems to improve response speed and support efficiency.
- Connected operational knowledge, inventory context, historical cases, and structured CRM actions within one governed AI-assisted workflow.

Audino – Text-to-SQL & SQL-to-Text AI Analytics Assistant**Alphabet / Google via Petra Pathway Consulting****Oct 2025–Feb 2026**

- Developed a generative-AI analytics engine that translated natural-language questions into optimized BigQuery SQL and converted query results into contextual business narratives.
- Built a FastAPI backend on Cloud Run with Gemini models, prompt routing, tool invocation, schema and context injection, response validation, and governed data access.
- Implemented Redis-backed sessions, caching, conversational memory, schema-aware follow-up handling, SQL safety guardrails, and explainability controls.
- Enabled governed self-service analytics while reducing dependency on manual SQL development and accelerating business insight generation.

Telecom CLTV, Churn & Segmentation Analytics Suite**LigaData Inc.****Jun 2025–Apr 2026**

- Designed and delivered an end-to-end telecom data-science suite for customer lifetime value, segmentation, churn prediction, revenue forecasting, and campaign optimization.
- Built scalable data preparation, feature engineering, model training, evaluation, experiment tracking, and batch-scoring pipelines using Python, Pandas, PySpark, Airflow, Redshift, SageMaker, XGBoost, scikit-learn, TensorFlow, and MLflow.
- Delivered Power BI, QuickSight, and Grafana dashboards exposing predictions, customer segments, behavioral trends, revenue opportunities, and operational KPIs.
- Integrated RAG and internal LLM capabilities to generate narrative explanations, automated labels, and business-friendly model insights.

Multilingual KYC/AML Name Screening System**Framework Solutions LLC****Feb 2025–Apr 2025**

- Engineered an AI-assisted financial name-screening system that aggregated, validated, and matched records from multiple compliance sources for automated KYC/AML checks.
- Built secure REST APIs for name submission, hashing, multilingual normalization, transliteration, alias handling, similarity scoring, and watchlist matching.
- Applied transformer-based language models to improve matching across spelling variations, multilingual formats, transliterations, and aliases.
- Designed a threshold-based decision engine to classify records as cleared or flagged for review, improving matching precision and reducing false-positive exposure.

Big Data Quality Control Suite
LigaData Inc.

Sep 2023–Jan 2025

- Solely designed and delivered an end-to-end data-quality platform spanning four data layers, from requirements and rule design through deployment, monitoring, and remediation.
- Built profiling, validation, reconciliation, and quality-scoring pipelines using Python, Pandas, PySpark, HDFS/DFS APIs, PostgreSQL, and Trino/Presto SQL.
- Enforced completeness, uniqueness, consistency, validity, timeliness, schema integrity, and operational-freshness metrics across production datasets.
- Automated duplicate detection, partially committed file reprocessing, issue classification, rule execution, and remediation workflows.
- Integrated Grafana dashboards and alerts, improving data quality by approximately 40%, reducing issue-resolution time by approximately 70%, and cutting investigation time by approximately 90%.

NYC Taxi Data Engineering & Analytics Platform
Independent Project

Apr 2024–Jun 2024

- Built a reusable pipeline to acquire public NYC taxi datasets from government sources, validate and transform the data, persist it in a transactional database, and prepare analytical datasets.
- Developed a Pandas-based database communication layer for extraction, transformation, querying, reusable analysis, and insight generation.
- Created Grafana dashboards to analyze operating patterns, answer business questions, and communicate trends to stakeholders.

HDFS/DFS Retention Policy Executor
LigaData Inc.

Jun 2023–Apr 2024

- Designed and deployed an automated data-retention engine across distributed file systems to enforce lifecycle and governance policies at scale.
- Built policy evaluation, data selection, execution, and audit workflows using Python, Pandas, PySpark, HDFS/DFS APIs, PostgreSQL, and Trino/Presto SQL.
- Containerized the solution with Docker and orchestrated execution on Kubernetes for scalable, repeatable, and controlled operation.
- Added complete audit tracking, Grafana dashboards, alerts, and operational controls, saving approximately 60% of developer and infrastructure time previously spent on retention work.

Figuroz Workforce Activity Analytics Tool
Independent Project

Oct 2022–Dec 2022

- Created a desktop data-collection application that captured keyboard, mouse, and application-usage events and uploaded records to a centralized online database in near real time.
- Built processing and dashboard capabilities for authorized companies to analyze utilization, activity patterns, time allocation, and productivity indicators.

Google AI4Code LLM/NLP Competition
Kaggle

Jun 2022–Aug 2022

- Engineered an NLP ranking model to infer relationships and ordering between code and markdown cells in Python notebooks.
- Combined DistilBERT representations with WaveNet-style gating to model code-comment relationships and notebook-cell sequences.
- Implemented parallel data loading, mixed-precision training, and optimizer improvements to accelerate experimentation and model iteration.

Medical Question-Answering NLP System Independent Project

Apr 2022–May 2022

- Built a medical question-answering proof of concept using BERT, DistilBERT, and Hugging Face Transformers on a clinic-provided dataset.
- Developed text preprocessing, model training, evaluation, and inference workflows for answering patient questions from domain-specific clinical content.
- Delivered reproducible source code through GitHub to support testing, maintenance, and future extension.

COVID-19 Field Data Collection Platform KADDB

May 2020–Jun 2020

- Built three Android applications for collecting, validating, synchronizing, and updating COVID-19 medical-status data in field workflows.
- Integrated Firebase for real-time data processing and implemented QR scanning, patient-status updates, record modification, and controlled deletion.

COVID-19 SEIQR Simulation & Forecasting Model KADDB

Apr 2020–Jun 2020

- Developed an SEIQR epidemiological simulation for infection-trend forecasting, scenario analysis, and policy-impact assessment.
- Built model-input and result-processing workflows and deployed outputs through a Django web application for public-health decision support.

Health Care Data Management System – Python / PostgreSQL StepByStep Academy

Nov 2019–May 2020

- Built a multi-role desktop health-care application for patient, doctor, administrator, and manager workflows with structured PostgreSQL data storage.
- Implemented registration, patient tracking, access workflows, validation, and reliable multi-user database interactions.

Health Care Data Management System – JavaFX / MS Access The Hashemite University

Sep 2017–Jan 2018

- Created a JavaFX desktop system for patient registration, tracking, database interaction, and operational health-care workflows.
- Applied object-oriented design and structured data-access patterns across application logic, user interfaces, and database operations.

Professional Development

- **Continuous Applied Development:** Maintain technical depth through targeted learning, hands-on experimentation, technical documentation, and production-oriented projects across AI, data science, cloud platforms, MLOps, and project delivery.

Volunteer Experience

Holding Hands Volunteer SOPRO – Portugal

Nov 2022–Feb 2023

- Designed and delivered school-based sessions on cultural understanding, human rights, social equality, inclusion, and civic awareness in Barcelos, Portugal.
- Coordinated with international volunteers and local educators to plan workshops, facilitate group activities, and promote cross-cultural learning.
- Strengthened public speaking, stakeholder coordination, facilitation, and communication across diverse audiences and education settings.

Taking Care of the Caretakers ADRA Polska – Poland

Aug 2022

- Participated in an international program focused on caregiver well-being, stress management, psychosocial support, and inclusive community service.
- Collaborated with participants from Croatia, Italy, Jordan, Poland, Russia, and Ukraine, strengthening cross-cultural teamwork, facilitation, and human-centered communication.